

Part 1: Choosing the Right Visuals for Clarity and Impact

Selecting The Charts

In making this reflection, we explore the designs and implementation choices made for a global COVID-19 visualization dashboard. The activity in this module uses three levels of data exploration to help readers understand the cumulative death counts over time. These levels include having a high-level overview, mid-level comparisons, and low-level dynamic tools. Each tool helps turn a massive dataset into a clear and meaningful story. Displaying raw information in a text-only format fails to convey patterns effectively (zyBooks, 2026). Using different deliberate chart structures instead allows viewers to interpret trends easily.

Using a Heatmap to Spot Patterns Quickly

First, a static heatmap was created using Python's Seaborn library to provide a high-level overview of the data. The original data set contains hundreds of rows and columns, which can easily confuse a reader. A heatmap solves this problem by condensing the large dataset into a simple grid. This kind of choice allows readers to instantly spot strong statistical patterns without straining their eyes. Using deliberate color variations helps direct the viewer's attention to the most important statistics right away (GeeksforGeeks, 2023).

Using a Bar Chart to Compare Different Groups

Next, a hierarchical bar chart was built using Matplotlib and Seaborn to show mid-level comparisons. This chart organizes geographic data by magnitude using horizontal bars. Larger regional averages are grouped neatly above individual national entries. A distinct color-coding system separates these categories. The clear separation ensures that smaller country-level data points do not get lost next to massive continental numbers. Following proper color theory

guidelines ensures that different data groups look distinct and visually balanced (GeeksforGeeks, 2023).

Using a Line Chart and Labels to Tell a Story

Finally, a time-series line chart was constructed to track cumulative deaths over time. Line charts are ideal for showing continuous changes over a period of time. To make the chart more impactful, clear text annotations were added directly to the line curve. These text callouts mark major historical events, such as the initial spikes in 2020 and the start of global vaccination campaigns. Adding direct annotations to a line graph gives readers important real-world context and removes the need to constantly check a separate legend (zyBooks, 2026).

Part 2: Overcoming Challenges with Interactive Charts and Animations

Fixing Slow Loading Times

Building interactive charts with large datasets often presents some of its own technical challenges. The raw file for this activity contained nearly half a million data points. Loading this much information at once caused the web browser to slow down and lag during filtering. To fix this bottleneck, specific filters were applied to remove unnecessary data columns. In addition, a Python scaler tool was used to shrink the numbers to a standard scale between 0.0 and 1.0, and the values were rounded to four decimal places. This data reduction dramatically decreased the file size and made the dashboard load much faster. Making sure the underlying data types are properly formatted is a necessary step before building any clean visualization (Wins with Data, 2022).

Fixing Broken Animations and Memory Failures

Another major challenge occurred when animating the data over time using Matplotlib's FuncAnimation tool. The final animation file was 21 megabytes, which exceeded the default memory limits of the Jupyter system and caused a compilation failure. To solve this, the internal memory limit was raised to 30 megabytes within the code. The animation settings were also adjusted to clear old text from the screen as new frames were loaded. Finally, the video was converted into an interactive JavaScript format so it could render smoothly in a basic web browser.

Part 3: Reducing The Mental Effort and Creating an Inclusive Design

Designing a Layout That Isn't Overwhelming

Good dashboards must have a balance of detailed information along with a clean layout to avoid overwhelming the viewer. This activity had a simple design rule: show a broad overview first, let the user zoom and filter, and then provide deep details on demand. Achieving this requires the dashboard to pair a global trend graph with a secondary detailed graph. When a user clicks or zooms on the main graph, the secondary view automatically updates to show details for that specific selection. This setup keeps the initial screen simple and tidy.

Hiding Data Behind Hover Tooltips to Keep Charts Clean

To further reduce visual clutter, interactive hover tooltips were built into the plots. Instead of crowding the page with permanent text labels, specific numbers and dates only appear when a user moves their mouse over a datapoint. This feature keeps the charts clean while still giving readers access to exact metrics whenever they want them. Using interactive tooltips and legend

controls allows the viewer to absorb multiple layers of information without experiencing visual fatigue (Wins with Data, 2022).

Making the Charts Easy to Read for Everyone

Accessibility was another major priority during the design process. Relying only on color hues to show meaning can make a chart impossible to read for colorblind individuals. Solving this problem requires colorblind-friendly palettes with high-contrast brightness levels. This choice ensures that the bars and lines remain easy to read, even in grayscale. Selecting high-contrast light intensities prevents accessibility barriers (GeeksforGeeks 2023). Lastly, all charts include clear titles, large fonts, and visible gridlines to make the data accessible to every reader.

References

zyBooks. (2026). Data Literacy and Visualization (SNHU CS540). Module Six: Generating Dynamic Data Visualizations. <https://learn.zybooks.com/>

GeeksforGeeks. (2023). Color theory to improve your data visualizations. <https://www.geeksforgeeks.org/data-visualization/color-theory-to-improve-your-data-visualizations/>

Wins with Data. (2022, October 18). *Interactive data visualization in Python for beginners* [Video]. YouTube. https://www.youtube.com/watch?v=wV59_q76gh0